Maahir Rajesh Garg

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Education

National University of Singapore Aug 2023 – May 2027 (Expected)
B.Comp. (Honours), Computer Science; Second Major in Quantitative Finance | GPA 4.71 / 5.00 *Singapore*
– Minor in Data Engineering; AI Specialisation (Distinction). Coursework: Software Engineering, Algorithms, Databases, Networks.

Technical Skills

Programming: Python, TypeScript/JavaScript, Java, SQL, C/C++, Swift, R, Bash
Web & Backend: React, Next.js, FastAPI, Pydantic, SQLAlchemy, REST APIs, PostgreSQL, Supabase, Firebase Auth
AI & ML: agentic LLMs, tool-calling, LLM evaluation, Arize tracing, PyTorch, Hugging Face, scikit-learn, XGBoost
Systems & Data: Docker, AWS, Snowflake, Spark, pandas, pytest, Playwright, GitHub Actions, Vercel, Git

Experience

AI Centre for Educational Technologies (AICET), Team Koditsu Aug 2026 – Present
AI Engineer | Python, pandas, scikit-learn, XGBoost, Quarto, LLM Evaluation *Singapore*
– Researching whether code-editing behaviour can flag likely cheating for human review in Koditsu's take-home coding assessments.
– Using real assessment attempts, trained a decision tree on the signals the production rule already uses and scored both on a held-out fifth of candidates, with validation folds split by candidate.
– Built the first Python codebase (**856 tests**) comparing logistic regression, XGBoost, rule-based baselines and an LLM judge.

GIC Jan 2026 – Jul 2026
AI Engineer | Python, Agentic LLMs, Arize, Chainlit, Snowflake, Docker/AWS *Singapore*
– Built an agentic CLI and Chainlit assistant over proprietary LLMs in a sandboxed Docker/AWS environment, with **8 custom tools**, multi-agent orchestration and skill-routing, for an **11-person team** whose data rules exclude tools such as Copilot.
– Designed an audit pipeline that calls Snowflake and SharePoint as tools and writes audit-ready reports from plain-English prompts, automating **2 manual data sources** and cutting evidence retrieval from **hours to near-instant**.
– Automated **50+ deterministic audit tests** and wrote hook-based guardrails that check every agent action.
– Added custom agent evaluations and Arize tracing so each model and tool call can be inspected for audit and debugging.

Interactive 3D Lab, in collaboration with Apple May 2025 – Dec 2025
Swift Developer, Apple Vision Pro | Swift, RealityKit, visionOS, ARKit *Singapore*
– Co-developed a Vision Pro and iPhone hand-eye tracking system for stroke-rehabilitation assessment at **89% cross-device accuracy**.
– Built the visionOS UI, hand and gaze tracking, and iPhone companion app at **30 fps**; tested with healthcare practitioners over 3 sessions of 3 patients each. Co-inventor on a patent application in progress.

Accelerice, NUS Overseas Colleges May 2024 – Aug 2024
Data Engineer | Python, APIs, Databases, Web Performance *Jakarta, Indonesia*
– Built Python/API dashboards and backend databases that improved operations-team data access by **40%**.
– Cut page-load time **25%** and raised organic traffic **20%** through performance and SEO work.

Projects

Trader's Edge (FYP, team of 2) | *FastAPI, React, PostgreSQL* | traders-edge-v2.vercel.app May 2026 – Present
– Built the engine of a web game that teaches beginners to trade over **40 levels**, each ending in a mission against rule-based rivals.
– Wrote the seeded market model, the order engine (market, limit and stop-loss orders with fees and spread) and 8 rival strategies; the server replays each match from its seed and stored orders, so future prices never reach the browser.
– Deployed on Render, Vercel and Neon behind 4 CI workflows running **835 tests** and a 50-seed game-balance check.

Linguini (CS3216, team of 4) | *React, FastAPI, Supabase* | learnwithlinguini.com Oct 2026 – Present
– Building a web app that turns a learner's photo into vocabulary, grammar and I-Spy lessons in Spanish, French, German or Korean.
– Built the multiplayer side: saved friends, asynchronous photo challenges and persistent private I-Spy chats (API and UI).
– Added the no-sign-up guest demo, stripped EXIF metadata from uploads, and stopped retries from storing duplicate photos.

PGPals (solo) | *Next.js, TypeScript, Supabase, PostgreSQL, Vercel* Jul 2026 – Sep 2026
– Built and ran the platform for a two-week residential challenge: **164 users** in **96 teams**, **2,297 submissions**, 3,763 media uploads.
– Enforced team access and deadlines in Postgres row-level security; 10 reviewers cleared every submission at a **3.2-hour** median.

Prediction-Market Research Engine (solo) | *Python, FastAPI, Pydantic, pytest* Feb 2026 – Jul 2026
– Built a read-only engine on live Polymarket and Kalshi data that prices cross-venue gaps after taker fees and order-book depth.
– Kept the pricing maths in pure functions apart from the venue adapters; **102 tests** cover contract matching, fees and calibration.

Honours

JEE top **0.5%** of 1.2M candidates; CBSE National Topper in Mathematics; NUS Honour List of Student Tutors (rated 4.8/5.0).